

1: Personal Computer

1.1 Anti-Virus

Viruses are the bane of many a PC user's existence. They can attack you through malicious web sites, corrupt important system files and even erase important data. However, you can keep yourself safe with anti-virus software. Anti-virus software comes in several types of packages – best of all, some of the most effective anti-virus tools are freeware. Virus detection can also be achieved through online checks. It is imperative to understand their necessity, especially since they can slow down system performance and prompt the user with unfamiliar commands.

Normally, anti-virus software has three modes of detection:

1. Signature based detection –

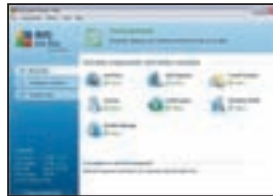
The most common method. It matches the contents of any infected files against a data base of virus signatures that are updated on a regular basis.

This mode even examines files in pieces to detect the

infected areas. However, constant updating of virus database definitions is essential.

2. Malicious activity detection – Monitors your system for suspicious behaviour. In case any suspicious activity is detected, the system initiates signature based detection. This is useful for catching unknown viruses that may not be listed in the database.

3. Heuristic based detection – The most system taxing mode of detection but the most effective for detecting unknown viruses. This is done through two methods: File analysis and file emulation. In file analysis, virus-like instructions that may be embedded in different programs rather than suspicious behaviour is investigated. If a program has instructions to format the C drive, for example, the anti-virus examines this program more closely. File emulation involves executing a program in a virtual environment and logging the actions that occur, according to which the anti-virus takes



AVG Free is one of the most reliable anti viruses around